



# **Code Craft**

**Innovate . Elevate . Inspire**

01



**Machine Learning**

02



**Deep Learning**

**A**

**Supervised**



**B**

**Unsupervised**



**C**

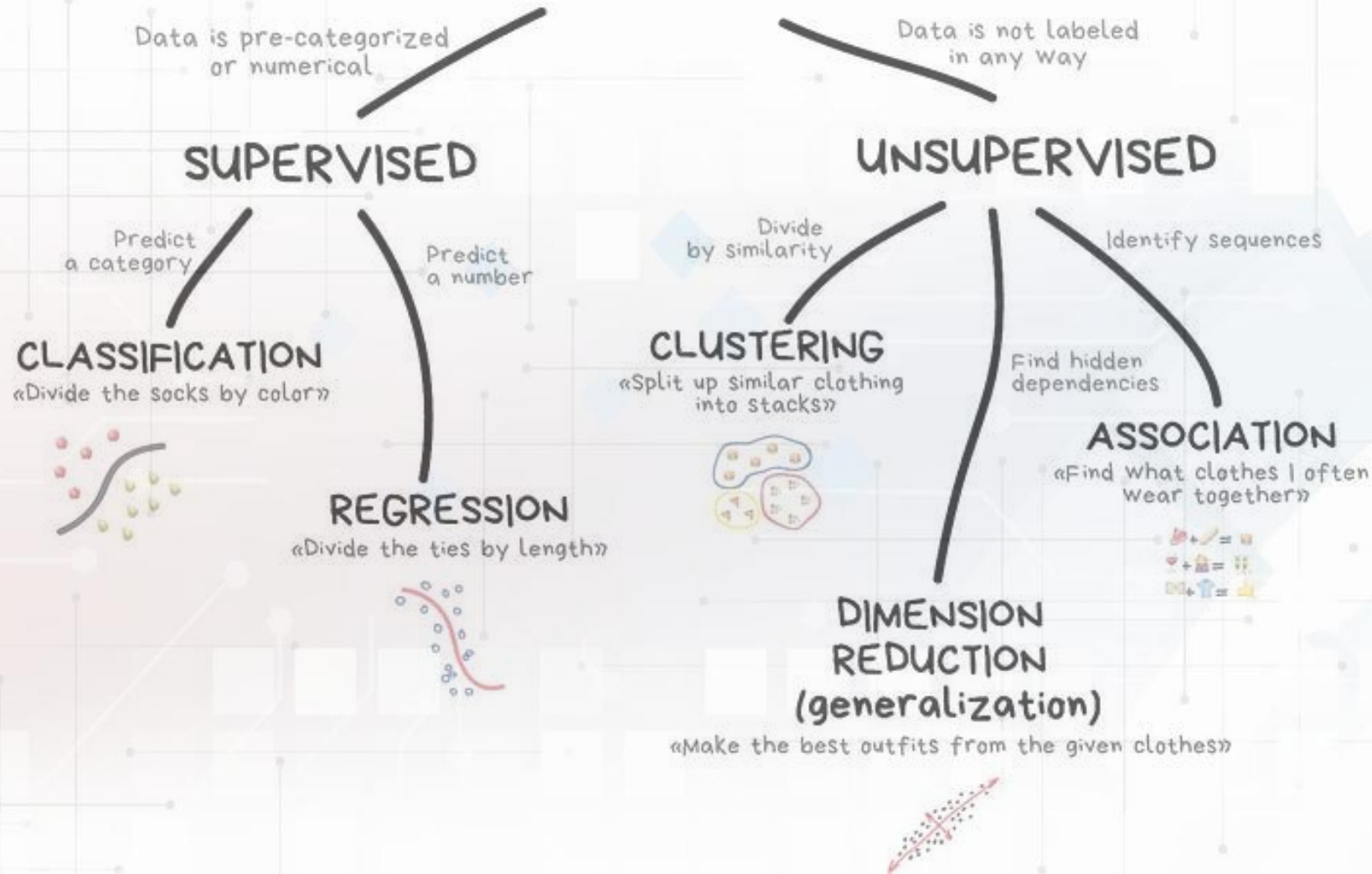
**Reinforcement**



**D Semi Supervised**

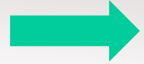
# Types of ML

# CLASSICAL MACHINE LEARNING



01

# Supervised



## Regression

- Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression

Maturity



## Classification

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- Naive Bayes

02

## Unsupervised



Clustering

- K-Means,
- Hierarchical
- DBSCAN



Association Rule

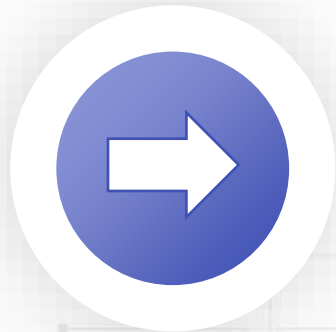
Association Rule

Anomaly Detection



Dimensionality  
Reduction

PCA (Principal Component Analysis)



# Linear Regression

is a supervised machine learning algorithm used to **predict a continuous numerical value** by finding a linear relationship between an input variable and an output variable.

In simple terms:

**It learns a straight-line relationship between input (X) and output (Y) and uses that relationship to make predictions.**

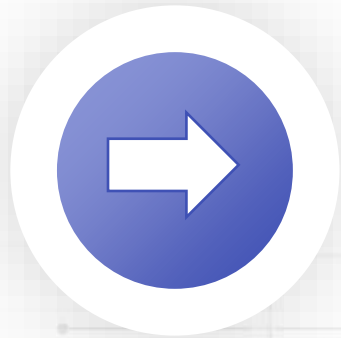
## Mathematical Formula

$$\hat{y} = mx + b$$

Where:

- $\hat{y}$  = Predicted output
- $x$  = Input feature
- $b$  = Intercept — predicted value when  $x = 0$
- $m$  = Slope/coefficient — how much Y changes when X increases by 1





## ➔ Linear Regression ( House price prediction)

It learns a straight-line relationship between input (X) and output (Y) and uses that relationship to make predictions.

### STEP 1: COLLECT DATA (AREA vs. PRICE)

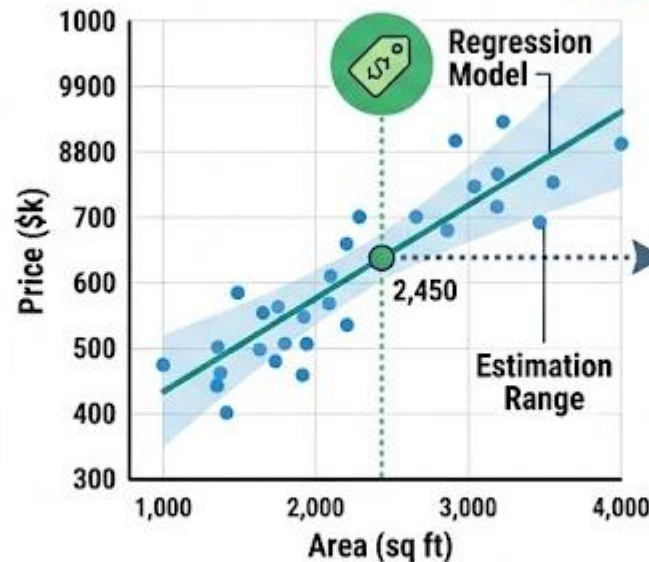
|                                                                                    | Area (sq ft) | Price (\$k) |
|------------------------------------------------------------------------------------|--------------|-------------|
|   | 1,500        | 400         |
|   | 2,000        | 580         |
|   | 2,500        | 680         |
|   | 3,000        | 850         |
|  | 3,500        | 950         |

#### New House to Predict:

Area: 2,450 sq ft



### STEP 2: FIND THE BEST FIT LINE (THE MODEL)



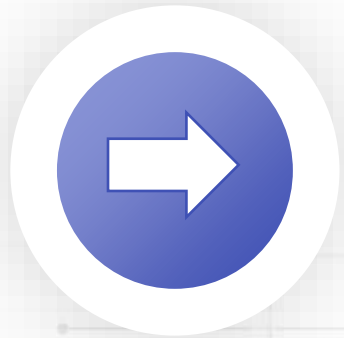
Model Type: Simple Linear Regression

### STEP 3: MAKE A PREDICTION

**PREDICTED  
HOUSE PRICE**

**\$685,000**

Estimation Range:  
**\$665k - \$705k**



# Logistic Regression

**Logistic Regression** is a supervised ML algorithm used to **predict the class/category of an input**.

## Mathematical Formula:

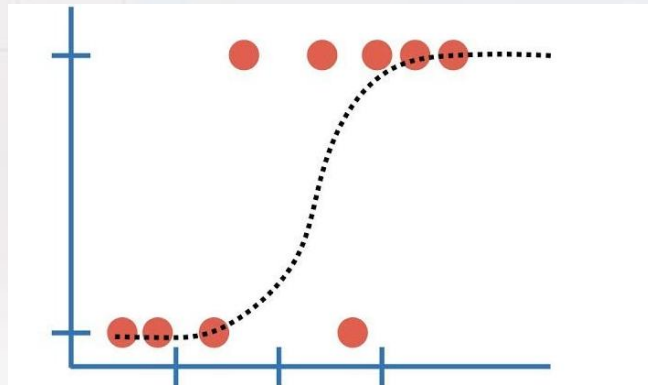
First, it calculates a linear combination:

$$z = b_0 + b_1x$$

Then it uses the **Sigmoid Function**:

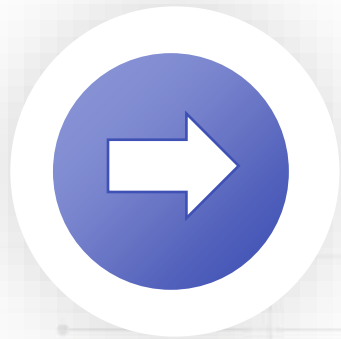
$$P(y = 1) = \frac{1}{1 + e^{-z}}$$

The output is between **0 and 1**.



## Use Cases

1. Spam Detection
2. Student Placement Prediction
3. Credit Card Fraud Detection
4. Medical Disease Diagnosis



# ➤ Logistic Regression

**Logistic Regression** is a supervised ML algorithm used to **predict the class/category of an input**.

## STEP 1: COLLECT DATA (CGPA & IQ vs. PLACED)

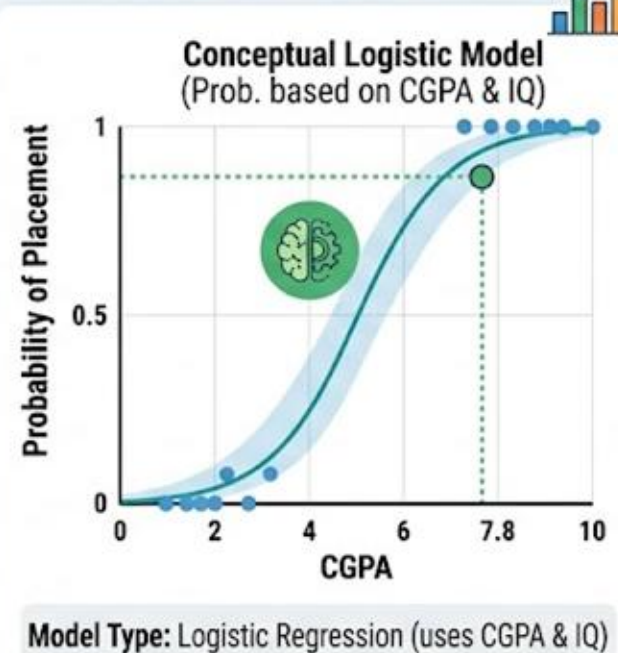
| Student | CGPA<br>(out of 10) | IQ Score | Placed?<br>(0=No, 1=Yes) |
|---------|---------------------|----------|--------------------------|
| S1      | 6.2                 | 90       | 0                        |
| S2      | 7.0                 | 110      | 1                        |
| S3      | 8.8                 | 125      | 1                        |
| S4      | 7.5                 | 100      | 0                        |
| S5      | 9.2                 | 130      | 1                        |

### New Student to Predict:

 CGPA: 7.8  
IQ Score: 115



## STEP 2: FIND THE LOGISTIC CURVE (THE MODEL)



## STEP 3: MAKE A PREDICTION

**PREDICTED  
PLACEMENT RESULT**

**82%**

Probability Range: 78% - 85%

Result: **Likely to be PLACED** 



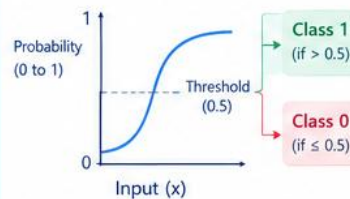
# ➔ Classification (Supervised)

## Logistic Regression

Predicts a class by calculating a probability and then applying a threshold.

How it works (simple)

$$P(y = 1) = \frac{1}{1 + e^{-(b_0 + b_1 x)}}$$



Example



### Spam Detection

Classify an email as Spam or Not Spam.

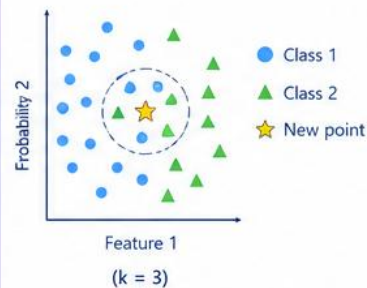
Key Takeaways

- 1 Predicts categories, not numbers.
- 2 Uses probability (Sigmoid function).
- 3 Simple and easy to interpret.

## K-Nearest Neighbors (KNN)

Classifies a new data point based on the majority class of its nearest neighbors.

How it works (simple)



### Location Recommendation

Recommend a location type based on nearby user preferences (e.g., restaurants, hotels).

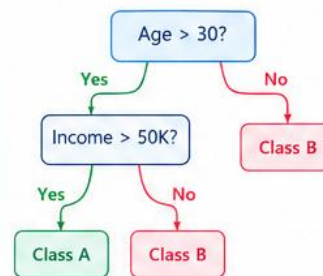
Key Takeaways

- 1 Works well with small datasets.
- 2 Uses similarity (distance).
- 3 Simple and intuitive.

## Decision Tree

Uses a series of simple questions (features) to split the data and make a decision.

How it works (simple)



Example

### Student Placement

Predict if a student will be placed or not based on CGPA, attendance, etc.

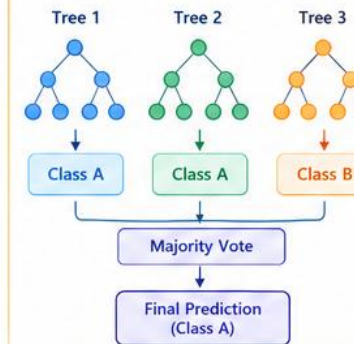
Key Takeaways

- 1 Easy to understand (rule-based).
- 2 Works with both numerical and categorical data.
- 3 Can overfit (use pruning).

## Random Forest

Builds multiple decision trees and combines their results (voting) for a final prediction.

How it works (simple)



### Example Disease Prediction

Predict the presence of a disease based on multiple health features (e.g., age, BP, cholesterol).

Key Takeaways

- 1 More accurate than a single tree.
- 2 Reduces overfitting.
- 3 Works well with large datasets.

## Naive Bayes

Uses Bayes' Theorem and assumes features are independent (of each other).

How it works (simple)

$$P(\text{Class} | \text{Features}) = \frac{P(\text{Features} | \text{Class}) \times P(\text{Class})}{P(\text{Features})}$$

| Feature         |   |     |
|-----------------|---|-----|
| Word: "offer"   | → | Yes |
| Word: "free"    | → | Yes |
| Word: "meeting" | → | No  |

Spam (Yes) / Not Spam (No)



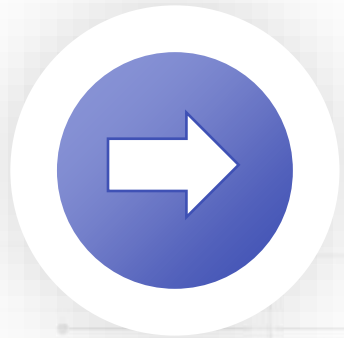
Example

### Email Classification

Classify an email as (e.g., 0, 1, 2,) Spam based on word frequencies.

Key Takeaways

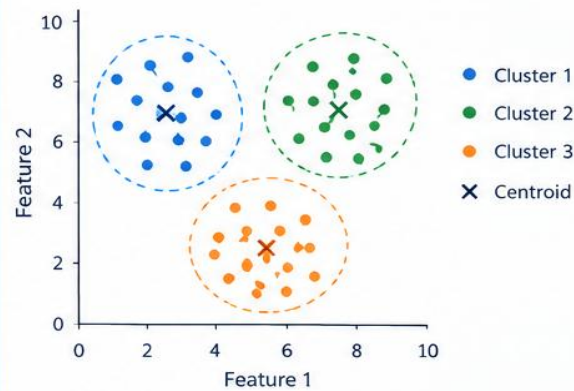
- 1 Fast and simple.
- 2 Works well with text data.
- 3 Assumes feature independence (a simplifying assumption).



# ➔ Unsupervised Learning

## K-Means Clustering

Groups similar data points into K clusters based on distance.

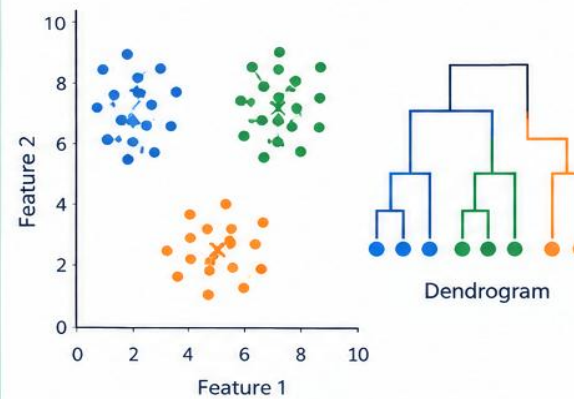


### Example

Customer segmentation  
Group customers with similar buying behavior.

## Hierarchical Clustering

Builds a tree-like structure (dendrogram) showing how data points are grouped.

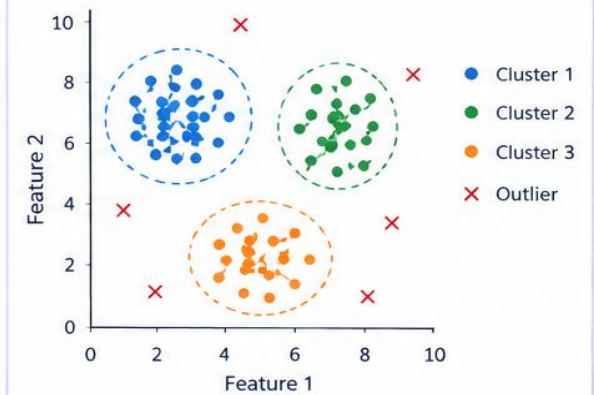


### Example

Document grouping  
Group similar text documents based on content.

## DBSCAN

Finds clusters based on high-density areas and detects outliers.



### Example

Fraud detection  
Identify normal transactions (clusters) and unusual ones (outliers).



# Anomaly Detection

Finds unusual or abnormal data points.



## Definition

Anomaly Detection is an unsupervised learning algorithm used to identify data points that are different from the normal pattern.



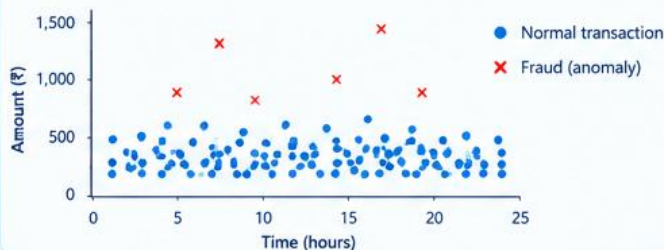
## How it works (in simple words)

- 1 Learns the normal pattern from data.
- 2 Checks new data against the pattern.
- 3 Flags the ones that look unusual as anomalies.



## Example 1 – Fraud Detection (Credit Card)

Most transactions are normal.  
A few are unusual (fraud).



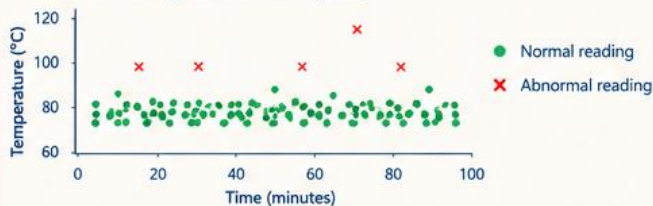
## Use Case

A bank uses anomaly detection to find suspicious transactions (like very high amount or unusual location).



## Example 2 – Machine Equipment Monitoring

Machine readings are usually stable.  
Some readings are abnormal (fault).



## Use Case

A factory uses it to detect machine faults before a breakdown happens.



## Key Takeaways

- 1 Finds unusual or abnormal data points.
- 2 Helps in fraud detection, fault prediction and safety.
- 3 Works well with normal vs. abnormal data patterns.



# Dimensionality Reduction

Reduces the number of features while keeping important information.



## Definition

Dimensionality Reduction is an unsupervised learning technique that transforms high-dimensional data into a lower number of features (dimensions) without losing important information.



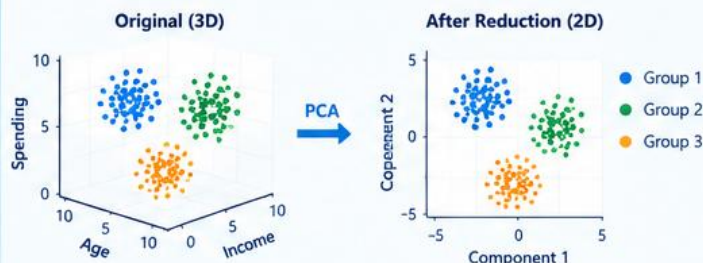
## How it works (in simple words)

- 1 Finds the most important patterns in the data.
- 2 Compresses the data into fewer features.
- 3 Helps in easier visualization and faster processing.



## Example 1 – PCA (Customer Data)

Original data has 3 features (e.g., age, income, spending).  
PCA reduces it to 2 features for easy visualization.



## Use Case

A retail company uses PCA to visualize customer segments in 2D and find patterns easily.



## Example 2 – Image Compression

A high-resolution image (many features) is reduced to fewer features using techniques like PCA or Autoencoders.



Original Image  
(High dimension)



Compressed Image  
(Low dimension)



## Use Case

Used in mobile apps and websites to store and display images more efficiently.



## Key Takeaways

- 1 Reduces the number of features (dimensions).
- 2 Keeps important information.
- 3 Helps in visualization, faster processing and better model performance.

```

! function (t, e) {
  "object" = typeof exports && "undefined" ≠ typeof module ? module.exports :
  "function" = typeof define && define.amd ? define(e) : (t = "undefined"
  globalThis ? globalThis : t || self).bootstrap = e()
}(this, (function () {
  "use strict";
  const t = {
    find: (t, e = document.documentElement) => [].concat(...Element.prototype.
    querySelectorAll.call(e, t)),
    findOne: (t, e = document.documentElement) => Element.prototype.quer
    (e, t),
    children: (t, e) => [].concat(...t.children).filter(t => t.matches(e)),
    parents(t, e) {
      const i = [];
      let n = t.parentNode;
      for (; n && n.nodeType === Node.ELEMENT_NODE && 3 ≠ n.nodeType;
      (e) && i.push(n), n = n.parentNode;

```

# Let's Code

[https://github.com/CodeCraftLearninghub/AI\\_Learning\\_exercises](https://github.com/CodeCraftLearninghub/AI_Learning_exercises)



# Deep Learning

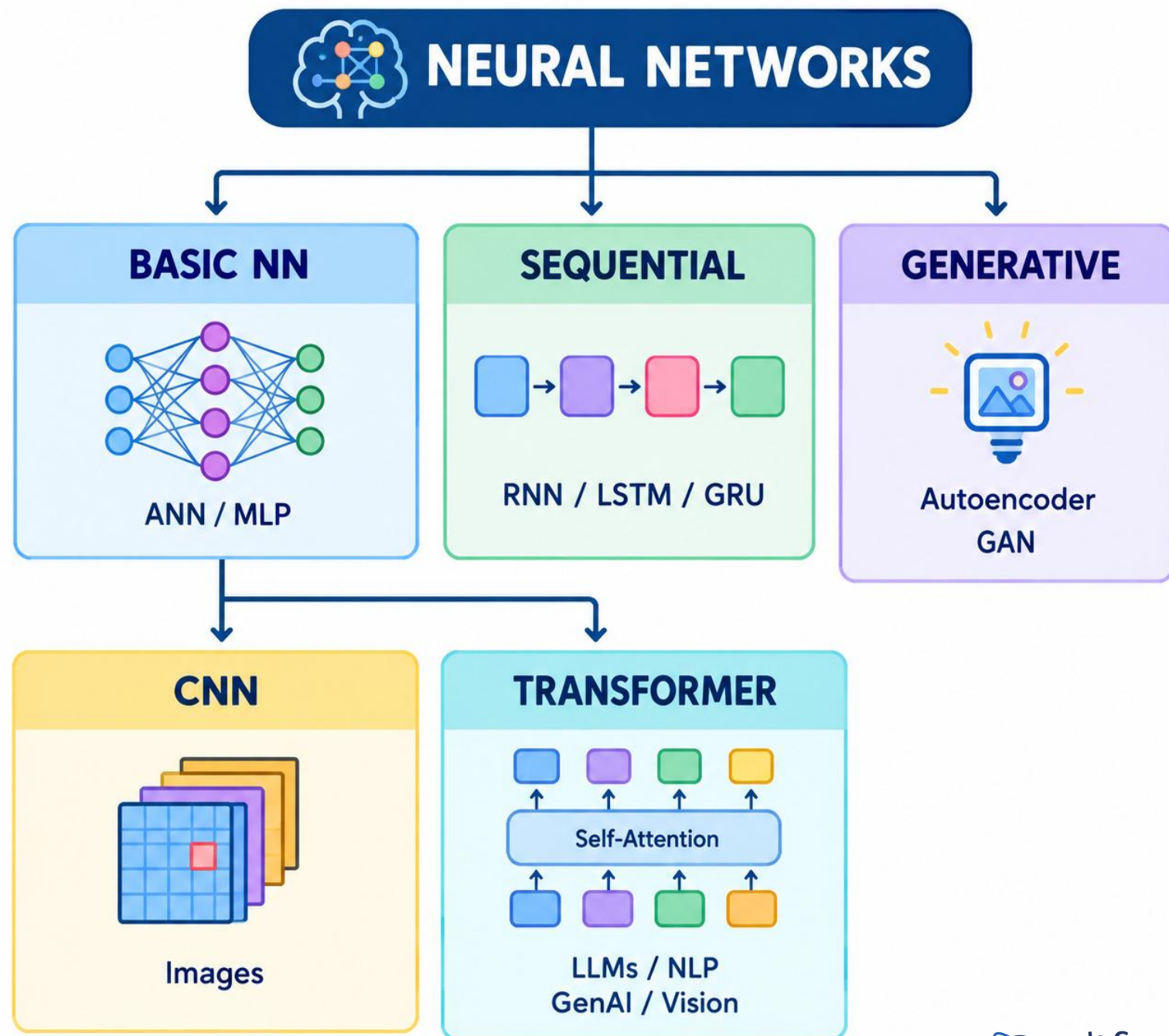
# What is Deep Learning ?

Is a subfield of Artificial intelligence and machine learning is inspired by the structure of a human brain.

Deep learning algorithms attempts to draw similar conclusions as humans would by continually analyzing the data with a given logical structure called neural networks

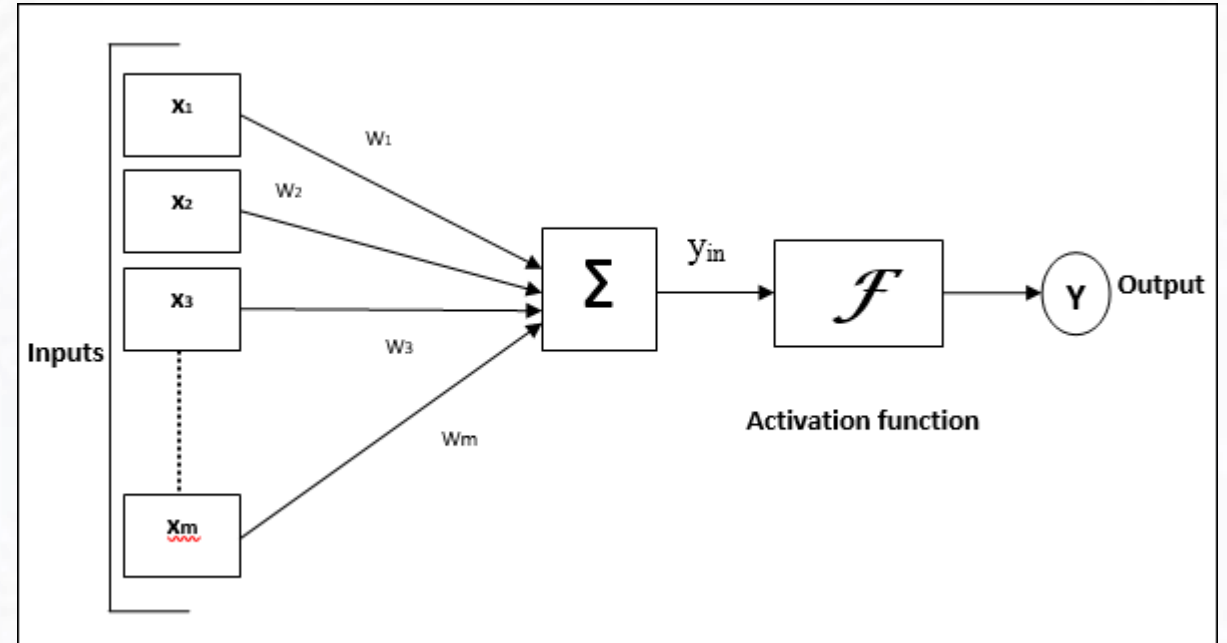
*ANN → CNN → RNN → LSTM → GRU → Autoencoder → GAN → Transformer*

# Types on NN



# Artificial Neural Network (ANN)

A neural network inspired by the human brain that learns patterns from data to make predictions.

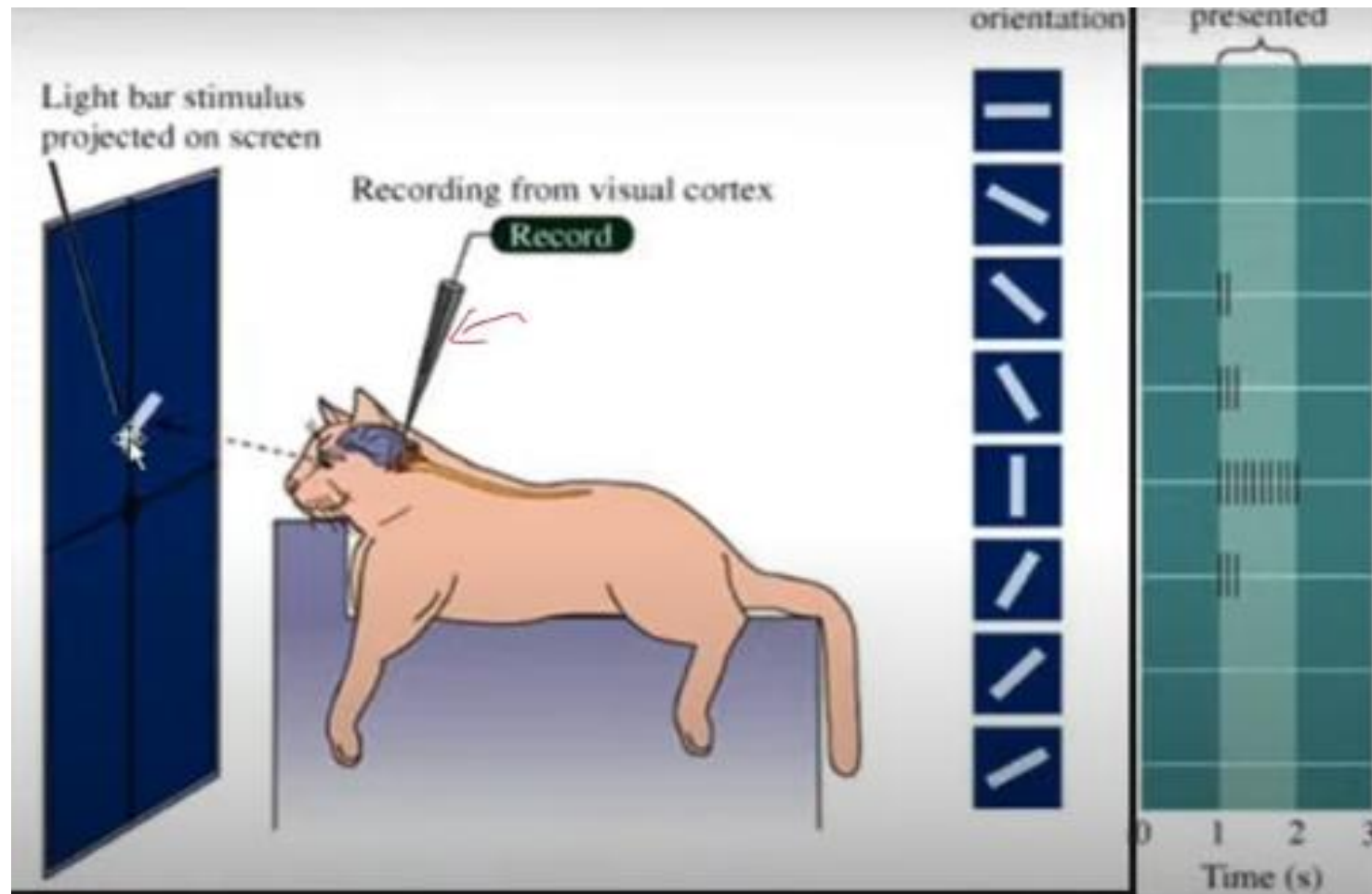


## Use cases:

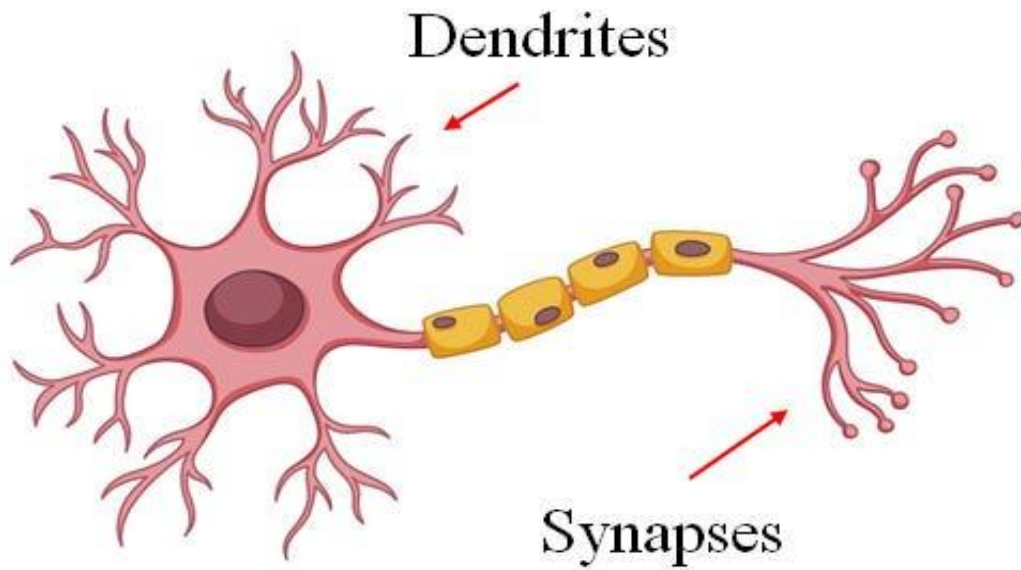
- 🎓 Student placement prediction
- 💳 Credit risk prediction

*ANN is the basic building block of many deep-learning systems.*

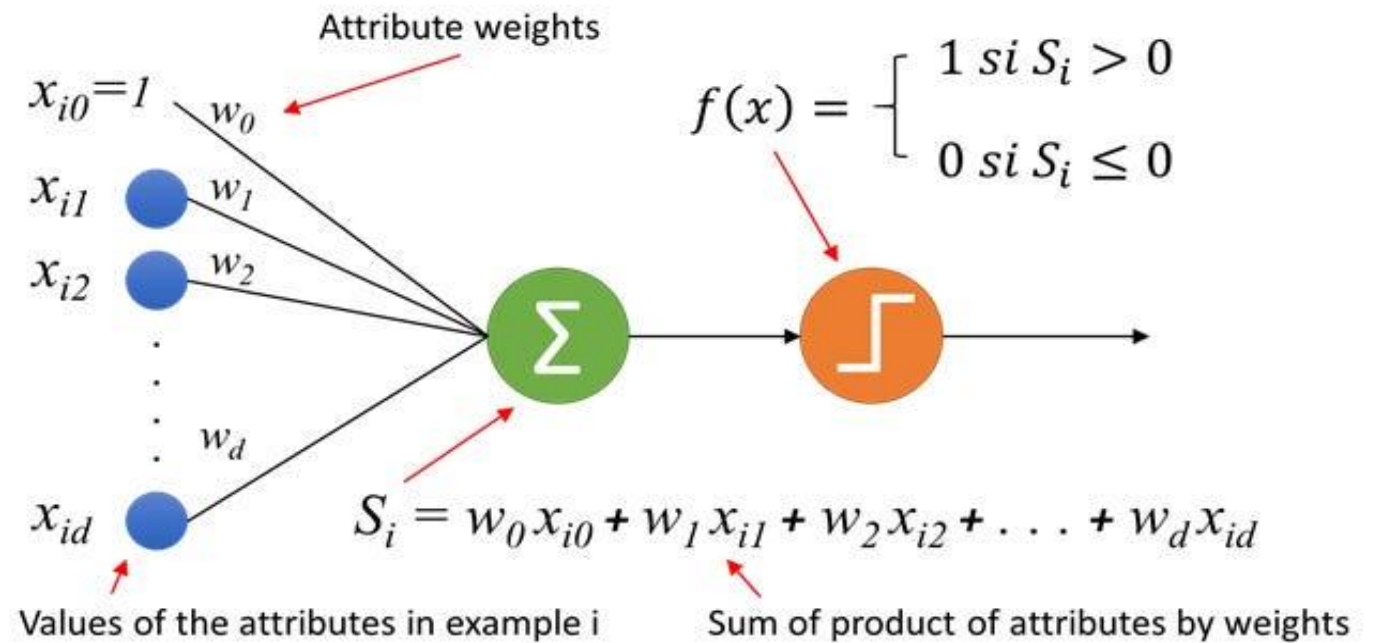
# Experiment



# Neuron & Perceptron

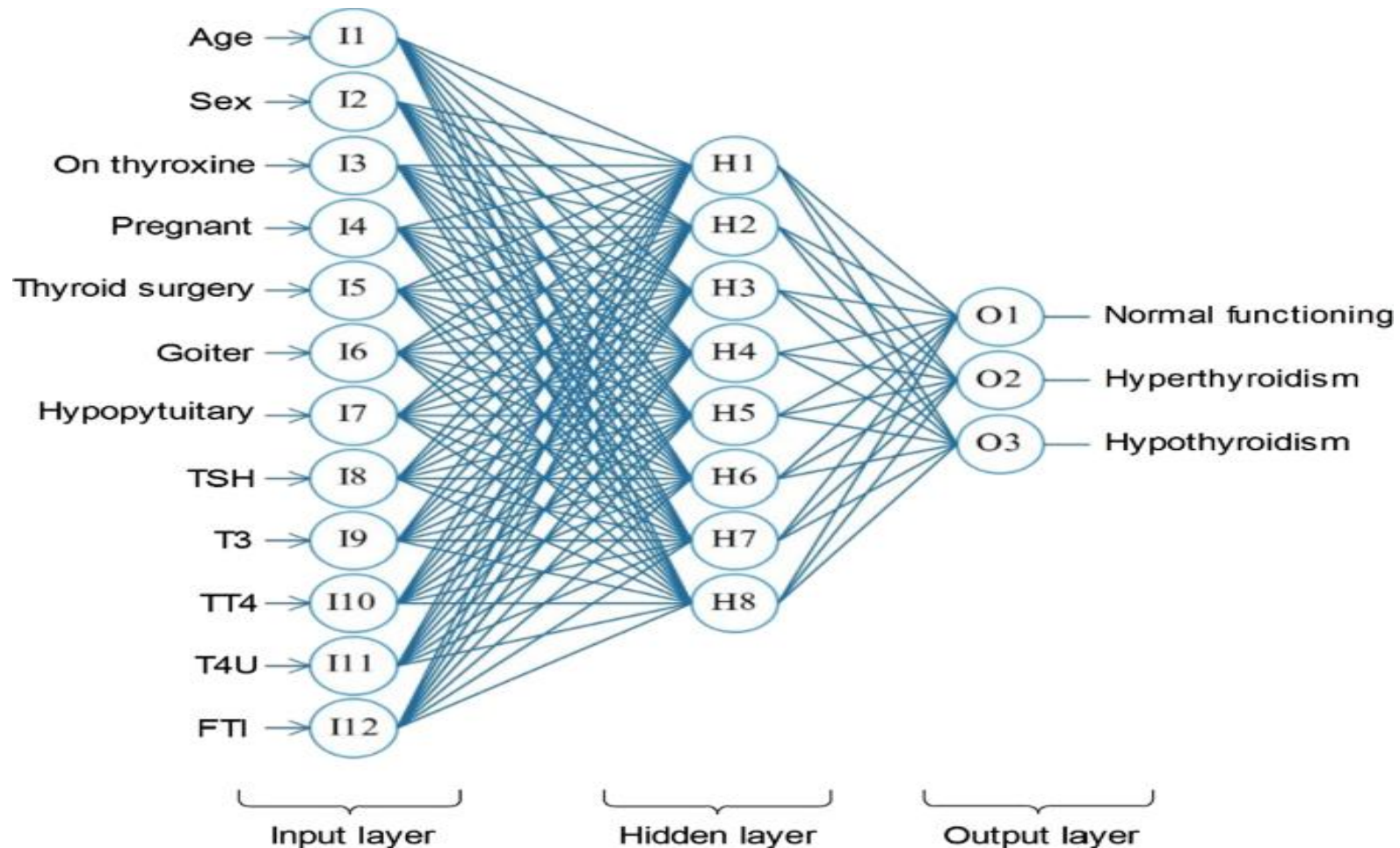


**NEURON**



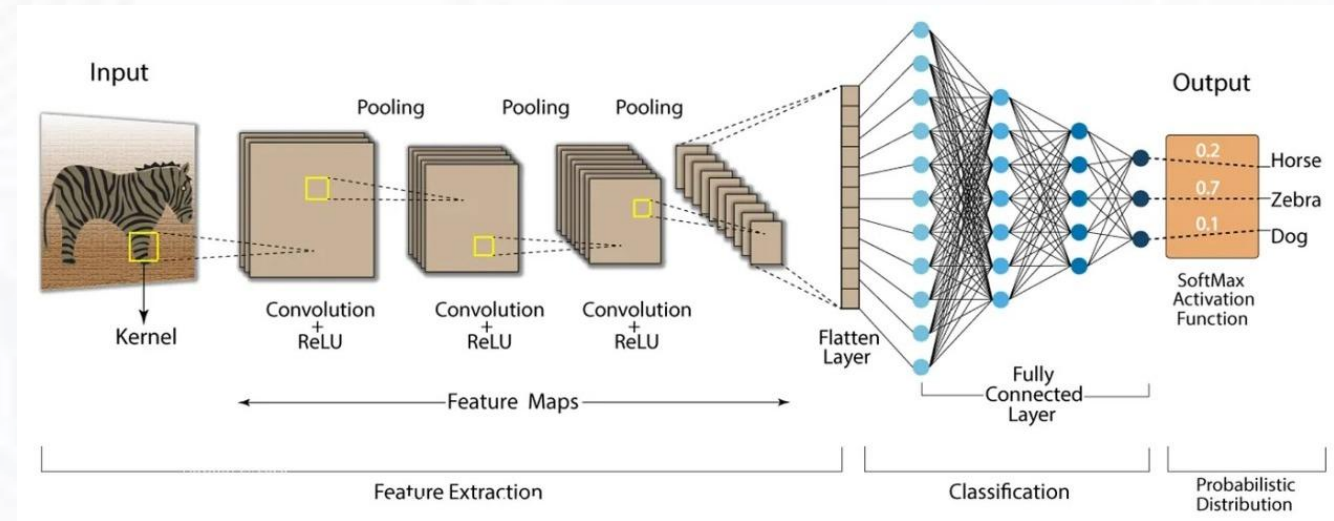
**PERCEPTRON**

# MLP (Multi Layer Perception)



# Convolutional Neural Network (CNN)

A neural network designed to understand images by detecting patterns such as edges, shapes and objects.

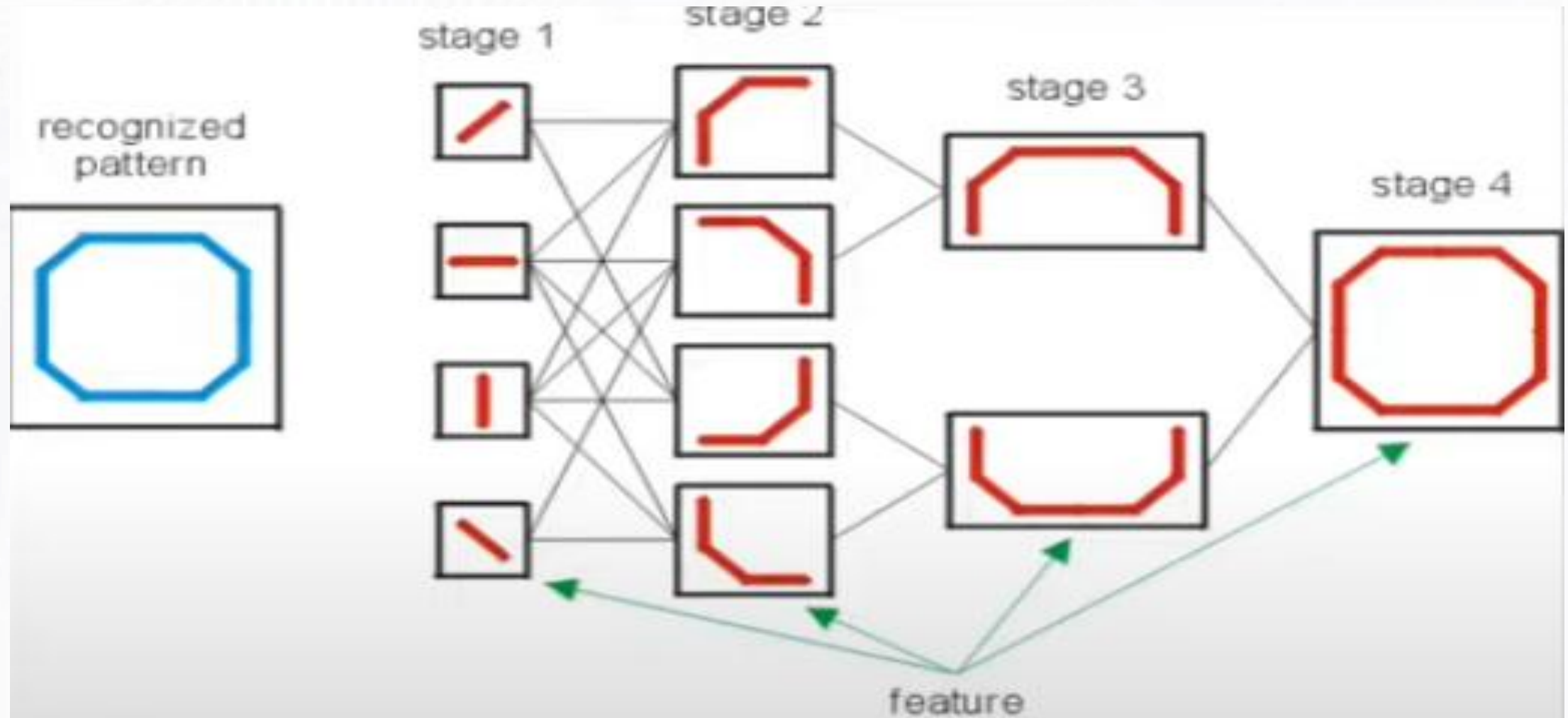


## Real-World Uses

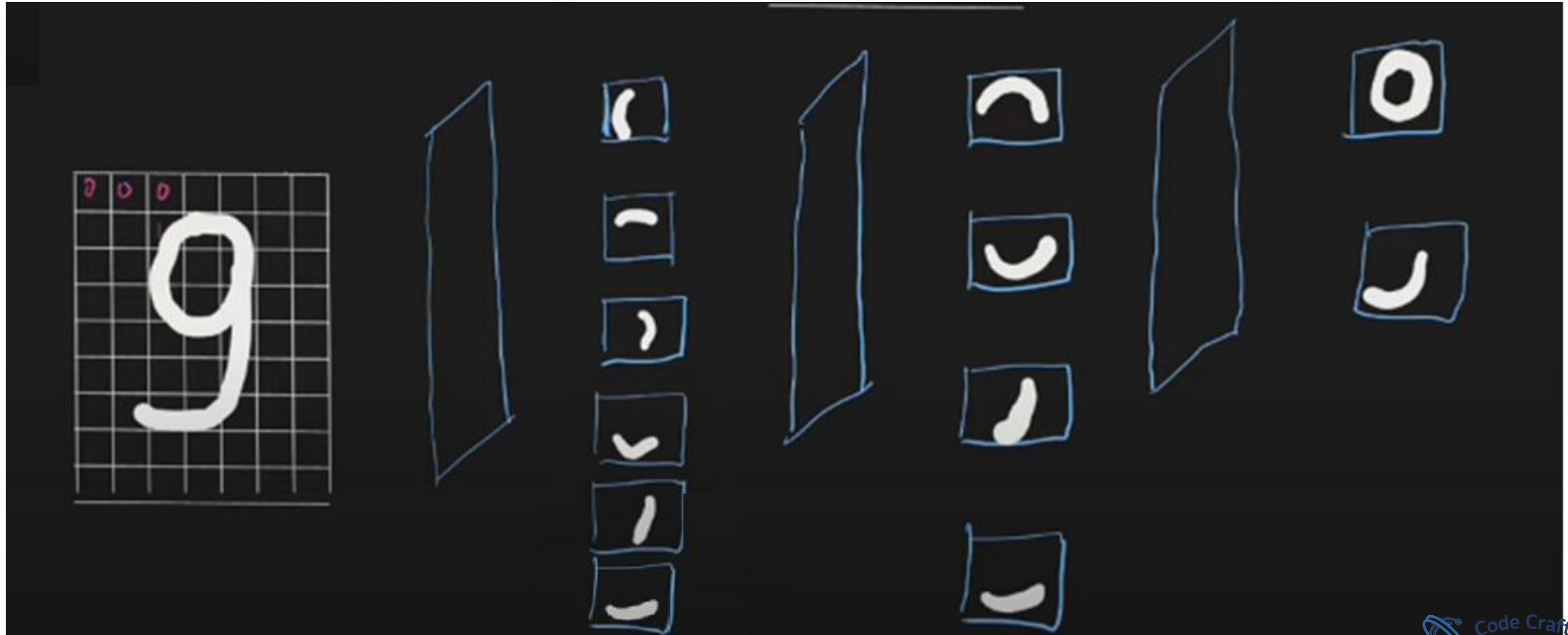
- ➔ **Self-driving vehicles** — detecting cars, pedestrians and road signs
- ➔ **Medical imaging** — identifying patterns in X-rays or scans

*CNN = Neural Network specialized for understanding images.*

# Convolutional Neural Network (CNN)

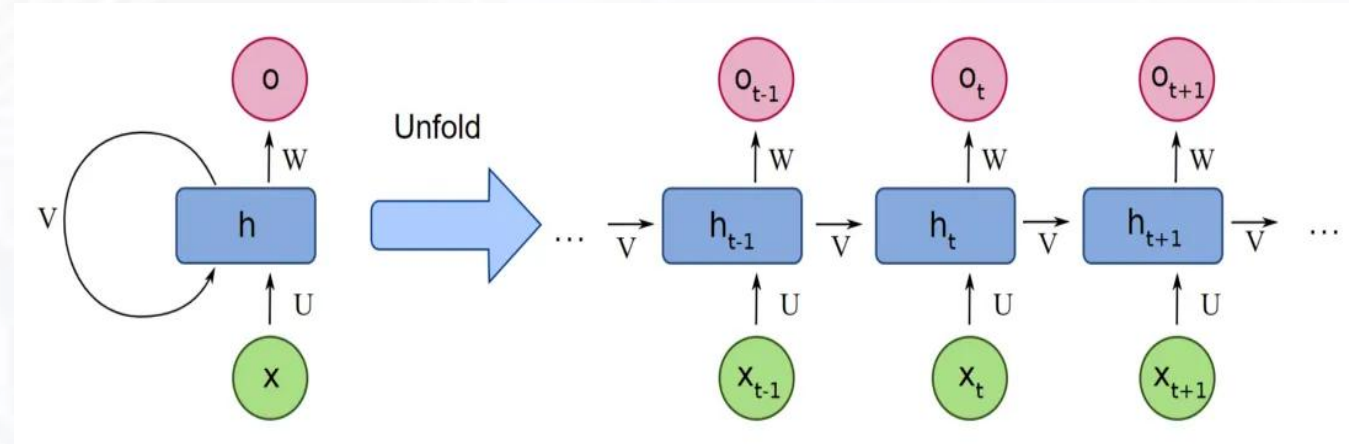


# Convolutional Neural Network (CNN)



# Recurrent Neural Network (RNN)

A neural network designed to work with sequential data, where previous information can influence the next prediction..



## Real-World Uses

- ➔ Speech recognition
- ➔ Time-series forecasting

*RNN is designed to understand sequences and order.*

# Generative Adversarial Network (GAN)

GAN uses two neural networks competing with each other to generate realistic new data.

## Real-World Uses

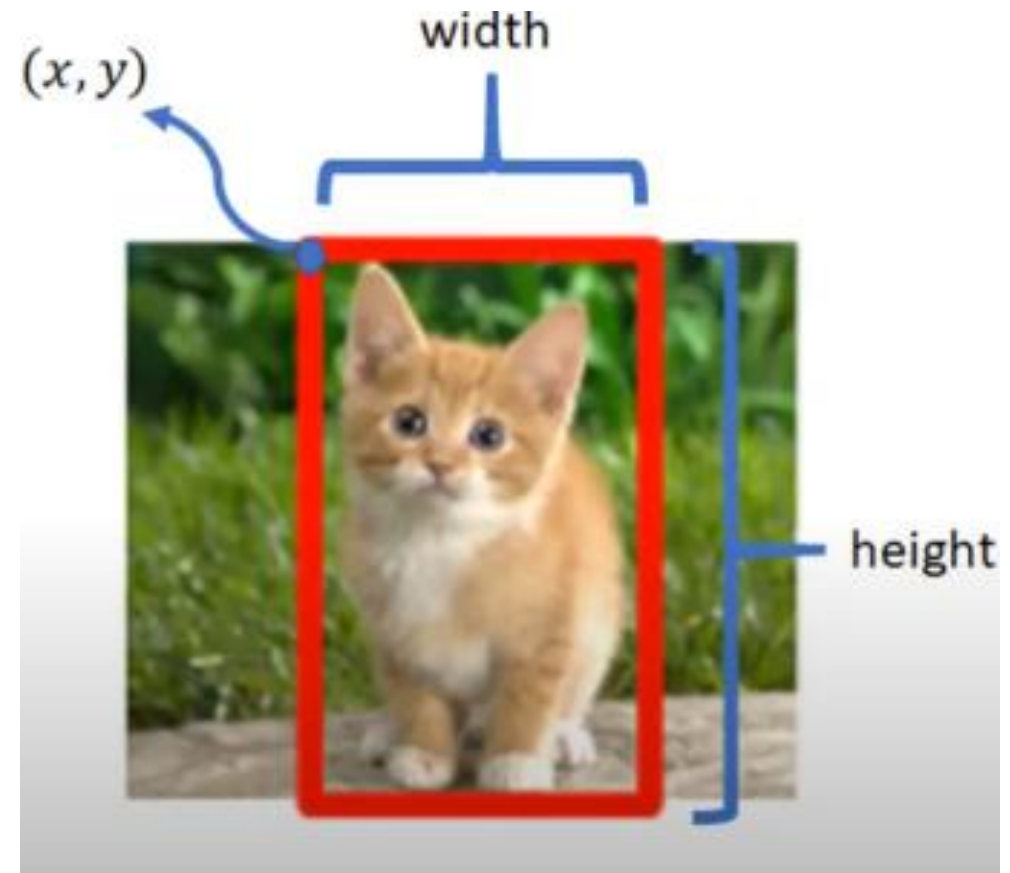
- ➔ Generating realistic images
- ➔ Creating synthetic training data



*GAN = Generator (creates) + Discriminator (checks).*

# Applications of CNN

Object Localization



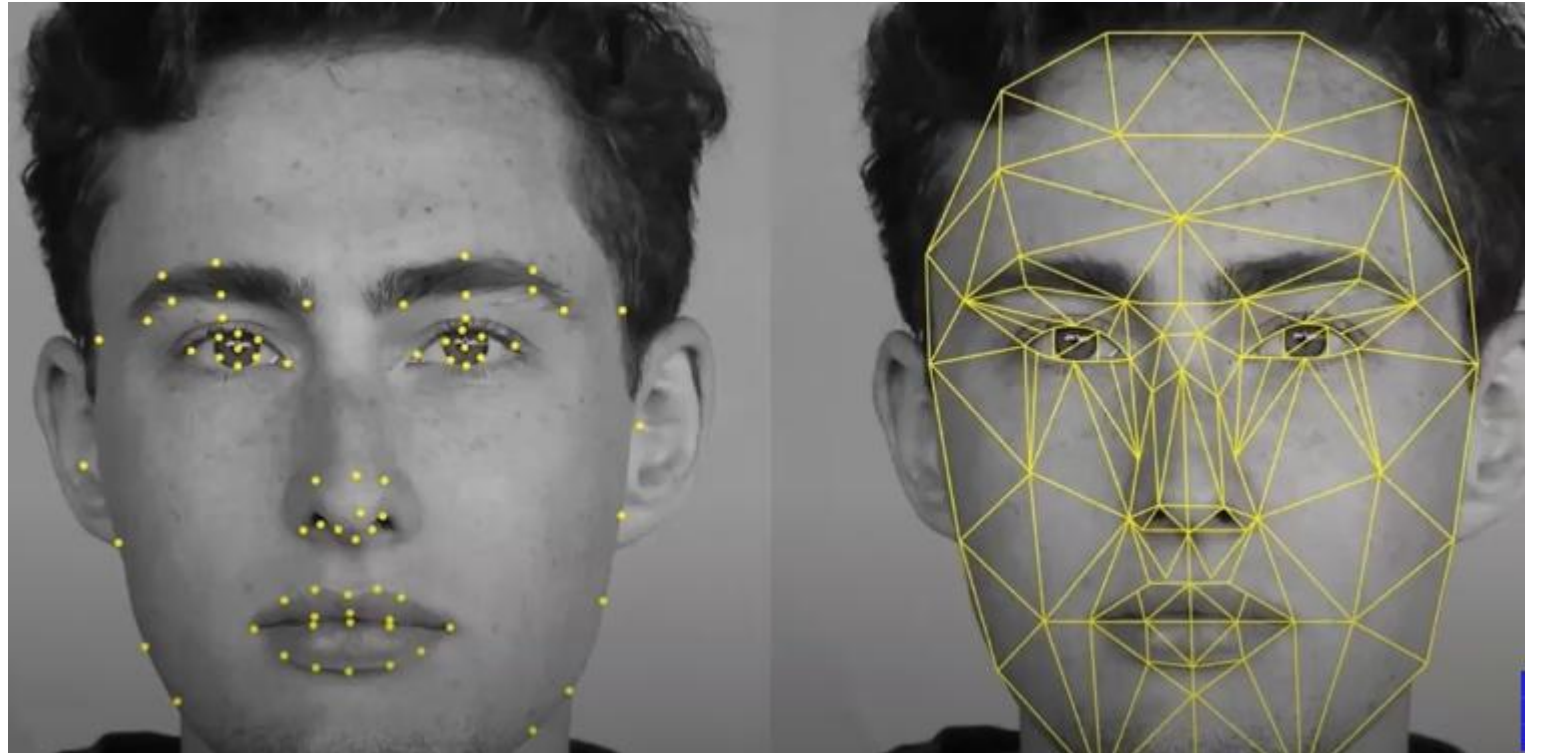
# Applications of CNN

## Object Detection Self Driving Cars



## Face Detection / Recognition

# Applications of CNN



# Applications of CNN

Image Segmentation



Image



Semantic Segmentation

# Applications of CNN

## Super Resolution

Low Resolution



High Resolution



# Applications of CNN

## Image Colorization



# Applications of CNN

Pose Estimation



```

! function (t, e) {
  "object" = typeof exports && "undefined" ≠ typeof module ? module.exports :
  "function" = typeof define && define.amd ? define(e) : (t = "undefined"
  globalThis ? globalThis : t || self).bootstrap = e()
}(this, (function () {
  "use strict";
  const t = {
    find: (t, e = document.documentElement) => [].concat(...Element.prototype.
    querySelectorAll.call(e, t)),
    findOne: (t, e = document.documentElement) => Element.prototype.quer
    (e, t),
    children: (t, e) => [].concat(...t.children).filter(t => t.matches(e)),
    parents(t, e) {
      const i = [];
      let n = t.parentNode;
      for (; n && n.nodeType === Node.ELEMENT_NODE && 3 ≠ n.nodeType;
      (e) && i.push(n), n = n.parentNode;

```

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